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COURSE CODE: CSA17

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Duration: 1 Hour
Total Marks:50

Scenario Based Assignment

I understand that any violation of academic integrity rules will result in disciplinary action.

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extremely large, weather patterns change dynamically, and the system may get stuck in locally optimal valve configurations.

Tasks:

Formulate this as an optimization problem and explain why traditional search is inefficient.

- Explain how Hill Climbing would work in this scenario and identify its limitations.
- Discuss issues like local maxima, plateaus, and ridges with real-world interpretation.
- Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations.
- Recommend the best approach and justify your choice based on scalability and adaptability.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

$$20 + 20 + 25 + 20 + 15$$

= 97

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1) Autonomous Sutmarmine - online search agent :

(i) . The sutmarmine explores an unknown trench where the complete map is unavailable .

- Online search is suitable because it makes decisions using current sensor data and update its path as new information is discovered .

(ii) . Limited sensor range means the sutmarmine cannot observe the entire environment .

- Hazards such as thermal vents , underwater currents and rock formations may appear unexpectedly , requiring continuous adaptation .

(iii) . It collects data from sonar and sensors while moving .

- The internal map is updated continuously , storing discovered details , safe paths and environmental conditions for future decisions .

Feature	Uninformed	Informed	Online
Environmental knowledge	None	Uses heuristic knowledge	Learned during exploration
Map Availability	Known	Known	Unknown / partial
Adaptability	Low	medium	High
Real time on	No	Limited	Yes

Online search is the best approach because it handles uncertainty, adapts to newly discovered hazards, and enables eye navigation in real time.

2) Hospital Duty Roster as CSP:

(i) Variables:

Assignment of nurses to shifts for each day.

Domains:

Available nurses who can work a specific shift.

Constraints:

1. No nurse can work overlapping shifts.
2. ICU shifts must be assigned only to qualified nurses.
3. Night shifts require senior nurse supervision.
4. Leave requests and availability must be respected.
5. Part-time nurses have limited work hours.

(ii)

- Assign nurses to shifts on day one.
- If a constraint is violated, undo the assignment & try another option.
- Continue until a valid schedule is found.

- iii) • After assigning a move, remove invalid choices from remaining shifts.
- Detect conflicts early and reduce unnecessary search.
- Improve efficiency by shrinking the search space.

iv)

Challenges :-

Last minute leave requests.

Increasing number of nurses & wards

Balancing workload fairly.

Best Strategy :-

Use Backtracking + Constraint Propagation.

3) Chess AI Opponent :-

- i). Minimax Treats chess as a two player adversarial game.
 - ii) The AI (max player) tries to maximize its score, while the opponent tries to minimize it.
- iii)
- ~~Chess~~ Searching every possible future move is computationally expensive.
 - Time and memory requirements grow exponentially with search depth.

(IV) Evaluation score =

Material advantage

King safety

Piece mobility

Board control

Position structure

Typical piece values:

Pawn = 1

Knight = 3

Bishop = 3

Rook = 5

Queen = 9

(V) • The minimax with Alpha-Beta pruning.

• Limit search depth according to time

• Use a strong evaluation function.

• This provides good move quality while meeting real time constraints for casual chess players.